

FAKHRI DINAL MAULANA PUTRA

AI/ML Engineer

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PROFESSIONAL SUMMARY

AI/ML Engineer specializing in Python, PyTorch, Computer Vision, and Generative AI. Passionate about building end-to-end AI applications, including RAG systems, computer vision models, and backend APIs. Experienced in designing modular, testable codebases and scalable architectures designed for deployment.

PROJECT EXPERIENCE

Knowledge Assistant (RAG System) — Project Lead & Core Developer

- Designed and developed a modular, end-to-end RAG system utilizing a FastAPI backend and Clean Architecture to enforce separation of concerns across API, Service, Core, and Database layers.
- Engineered a modular document processing pipeline supporting PDF, DOCX, TXT, and Markdown files, generating clean text chunks and indexing them in a ChromaDB vector database.
- Implemented local semantic search using HuggingFace sentence-transformers and integrated local LLMs (Llama 3 via Ollama) for multi-turn conversational Q&A with context pruning and conversational memory.
- Automated test coverage to ensure reliability of backend logic and containerized the system using Docker to make it designed for deployment.

Vegetable Object Detection & Classification System — Core Deep Learning Engineer

- Developed an object localization and classification system in PyTorch using a ResNet18 backbone to identify multiple vegetable classes and predict bounding box coordinates.
- Utilized NumPy and Pandas for data preprocessing, augmentation, and model training routines with cross-entropy loss for classification and MSELoss for coordinate regression.
- Created an interactive, real-time web application using Streamlit to handle image uploads, execute inference, and dynamically draw predicted bounding boxes on the UI.
- Curated a custom image dataset using Raspberry Pi and Pi Camera, configuring labeling structures and evaluation reports using Scikit-Learn.

Collaborative AI Agent Orchestrator — Lead Backend & Multi-Agent Systems Engineer

- Designed and built a production-ready, asynchronous multi-agent orchestration platform using CrewAI and LangChain, structured under Clean Architecture principles (Core, Services, API, Schemas) using a FastAPI backend.
- Developed an in-memory task repository with background ThreadPoolExecutor execution to manage task lifecycle states (PENDING, RUNNING, COMPLETED, FAILED) asynchronously and capture incremental agent execution logs in real-time.
- Integrated local LLM inference using Ollama (Llama 3/3.2) with customized agent profiles, task descriptions, and callback functions, implementing automated telemetry bypasses and console rendering deadlock resolutions.
- Created an interactive, real-time web application using Streamlit featuring API health checking, status polling, live markdown report rendering, and file export options.
- Containerized the services using Docker and orchestrated multi-container deployment via Docker Compose, configuring network communications to bridge containerized backend calls with host-level LLM services.

EDUCATION

Bachelor of Computer Science

Siliwangi University

Grade: 3.60

SKILLS

- AI/ML & Generative AI:** Large Language Models (LLMs), Multi-Agent Systems, CrewAI, RAG pipelines, Embeddings, Vector Databases, Transformers, YOLO11, OpenCV, PyTorch, TensorFlow, ResNet18
- Backend & Frameworks:** Python, FastAPI, Flask, Clean Architecture, Streamlit
- Databases & Storage:** ChromaDB, MongoDB
- Data Science & Tools:** NumPy, Pandas, Scikit-Learn (sklearn), Matplotlib, Docker, CI/CD, Git, Ollama, HuggingFace